PADHAKU
Science · Class 10 Science



CHAPTER 1

Chemical Reactions and Equations

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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⭐ Activities · Introduction

Activity 1.1 **Burn a cleaned magnesium ribbon (3–4 cm) held with tongs and collect the ash in a watch-glass.**

What do you observe?

✔ ANSWER

Observation: The ribbon burns with a **dazzling white flame** and changes into a **white powder** (ash).

Conclusion: The white powder is **magnesium oxide**, formed when magnesium reacts with oxygen of the air. A new substance is formed, so a chemical reaction has taken place.

$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

Activity 1.2 **Take lead nitrate solution in a test tube. Add potassium iodide solution to it.**

What do you observe?

✔ ANSWER

Observation: A **yellow precipitate** is formed.

Conclusion: The yellow solid is **lead iodide** (PbI_2). A change in colour/state shows a chemical reaction.

$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$

Activity 1.3 **Add dilute hydrochloric acid or sulphuric acid to a few zinc granules in**

a conical flask/test tube.

Do you observe anything happening around the zinc granules?

Touch the flask. Is there any change in its temperature?

✔ ANSWER

Observation: Bubbles of a gas (hydrogen) form around the zinc granules.

The flask becomes **warm**: the temperature rises.

Conclusion: Evolution of a gas and change in temperature show that a chemical reaction has taken place.

$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$

⭐ 1.1 Chemical Equations · In-text

? **Using these steps, can you balance Eq. (1.2): $Mg + O_2 \rightarrow MgO$?**

✔ ANSWER

- O: 2 on LHS, 1 on RHS → write 2MgO.
- Mg: now 2 on RHS, 1 on LHS → write 2Mg.

$2Mg + O_2 \rightarrow 2MgO$

Check: Mg 2 = 2, O 2 = 2. Balanced.

❓ Questions · 1.1 Chemical Equations

Q 1 **Why should a magnesium ribbon be cleaned before burning in air?**

✔ ANSWER

Magnesium reacts with air and gets a layer of **magnesium oxide** (and other impurities) on its surface.

This layer stops the ribbon from burning easily. Rubbing with sandpaper removes it, so pure magnesium comes in contact with oxygen and burns properly.

Q 2 (i) **Write the balanced equation: Hydrogen + Chlorine → Hydrogen chloride**

✔ ANSWER

$H_2 + Cl_2 \rightarrow 2HCl$

Q 2 (ii) **Barium chloride + Aluminium sulphate → Barium sulphate + Aluminium chloride**

✔ ANSWER

$3BaCl_2 + Al_2(SO_4)_3 \rightarrow 3BaSO_4 + 2AlCl_3$

Q 2 (iii) **Sodium + Water → Sodium hydroxide + Hydrogen**

✔ ANSWER

$2Na + 2H_2O \rightarrow 2NaOH + H_2$

Q 3 (i) Write a balanced equation with state symbols: Solutions of barium chloride and sodium sulphate in water react to give insoluble barium sulphate and the solution of sodium chloride.

✔ ANSWER

$BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$

Q 3 (ii) Sodium hydroxide solution (in water) reacts with hydrochloric acid solution (in water) to produce sodium chloride solution and water.

✔ ANSWER

$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$

⭐ 1.2.1 Combination Reaction

Activity 1.4 Slowly add water to a small amount of calcium oxide (quick lime) in a beaker. Touch the beaker.

Do you feel any change in temperature?

✔ ANSWER

Observation: The beaker becomes **hot**; the reaction is vigorous (hissing sound).

Conclusion: Calcium oxide combines with water to form slaked lime, releasing a lot of heat. It is a **combination** and **exothermic** reaction.

$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$

? **Identify the type of the reaction taking place in Activity 1.1, where heat is given out along with the formation of a single product.**

✔ ANSWER

Burning of magnesium is a **combination reaction** (single product, MgO) and also an **exothermic reaction** (heat and light are given out).

$2Mg(s) + O_2(g) \rightarrow 2MgO(s) + Heat$

⭐ 1.2.2 Decomposition Reaction

Activity 1.5 **Heat about 2 g of ferrous sulphate crystals in a dry boiling tube.**
Note the colour of the crystals before and after heating.

✔ ANSWER

Observation: The **green** crystals first lose water and turn white, then a **reddish-brown** solid (ferric oxide) is left. A smell of burning sulphur is noticed.

Conclusion: A single reactant breaks into simpler products: **thermal decomposition**.

$2FeSO_4(s) \rightarrow (Heat)\ Fe_2O_3(s) + SO_2(g) + SO_3(g)$

Activity 1.6 **Heat about 2 g lead nitrate powder in a boiling tube held with tongs.**

What do you observe? Note down the change, if any.

✔ ANSWER

Observation: Brown fumes of nitrogen dioxide (NO_2) are given off; a yellowish solid (lead oxide) is left.

Conclusion: Lead nitrate undergoes **thermal decomposition**.

$2Pb(NO_3)_2(s) \rightarrow (Heat)\ 2PbO(s) + 4NO_2(g) + O_2(g)$

Activity 1.7 **Electrolysis of water (with a few drops of dilute sulphuric acid) using carbon electrodes and a 6 V battery; gases collected in two inverted test tubes.**

Is the volume of the gas collected the same in both the test tubes?

Bring a burning candle close to the mouth of each test tube. What happens in each case?

Which gas is present in each test tube?

✔ ANSWER

Volume: Not the same. One tube (at the negative electrode, cathode) collects **double** the volume of gas than the other (at the positive electrode, anode).

Candle test: In the tube with more gas, the gas burns with a **'pop' sound** → **hydrogen**. In the other tube, the candle burns **more brightly** → **oxygen**.

Conclusion: Electricity decomposes water into hydrogen and oxygen in the ratio 2 : 1 by volume (**electrolytic decomposition**).

$2H_2O(l) \rightarrow (Electricity)\ 2H_2(g) + O_2(g)$

Activity 1.8

Take about 2 g silver chloride in a china dish. What is its colour?

Place it in sunlight for some time and observe its colour.

✔ ANSWER

Colour before: white.

Observation: It turns **grey** in sunlight.

Conclusion: Light decomposes silver chloride into silver and chlorine (**photolytic decomposition**).

$2AgCl(s) \rightarrow (Sunlight)\ 2Ag(s) + Cl_2(g)$

? **What form of energy is causing these decomposition reactions (reactions 1.22 and 1.23)?**

✔ ANSWER

Light energy (sunlight) causes the decomposition of silver chloride and silver bromide.

Activity **Carry out the following Activity: Take about 2 g barium hydroxide in a test tube. Add 1 g of ammonium chloride and mix with a glass rod. Touch the bottom of the test tube with your palm.**

What do you feel? Is this an exothermic or endothermic reaction?

✔ ANSWER

The bottom of the test tube feels **cold**.

The reaction absorbs heat from the surroundings, so it is an **endothermic reaction**.

$Ba(OH)_2 + 2NH_4Cl \rightarrow BaCl_2 + 2NH_3 + 2H_2O$

? Questions · 1.2.1–1.2.2 Combination and Decomposition

Q 1 (i) **A solution of a substance 'X' is used for whitewashing. Name the substance 'X' and write its formula.**

✔ ANSWER

'X' is **calcium oxide** (quick lime), formula **CaO**.

(It reacts with water to give slaked lime, $Ca(OH)_2$, whose solution is used for whitewashing.)

Q 1 (ii) **Write the reaction of the substance 'X' named in (i) above with water.**

✔ ANSWER

$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$

(Quick lime) (Slaked lime)

Q 2 **Why is the amount of gas collected in one of the test tubes in Activity 1.7 double of the amount collected in the other? Name this gas.**

✔ ANSWER

Water (H_2O) contains hydrogen and oxygen in the ratio **2 : 1** by number of atoms.

On electrolysis, 2 molecules of water give 2 molecules of hydrogen and 1 molecule of oxygen: $2H_2O \rightarrow 2H_2 + O_2$.

So the volume of hydrogen is double that of oxygen. The gas collected in double amount is **hydrogen**.

⭐ 1.2.3 Displacement Reaction

Activity 1.9 **Dip two cleaned iron nails in copper sulphate solution (test tube B) for 20 minutes; keep tube A and one nail for comparison.**

Compare the blue colour of the solutions in A and B, and the colour of the nails.

✔ ANSWER

Observation: The blue colour of copper sulphate in tube B **fades** (becomes light green). The dipped iron nails get a **brownish (copper) coating**; the nail kept aside stays grey.

Conclusion: Iron is more reactive than copper and displaces copper from copper sulphate (**displacement reaction**).

$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$

? **Why does the iron nail become brownish in colour and the blue colour of copper sulphate solution fades?**

✔ ANSWER

Iron displaces copper from copper sulphate solution.

The copper set free deposits on the nail, making it **brownish**.

Blue copper sulphate changes into **pale green iron sulphate**, so the blue colour fades.

$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$

⭐ 1.2.4 Double Displacement Reaction

Activity 1.10 **Mix about 3 mL sodium sulphate solution with about 3 mL barium chloride solution.**
What do you observe?

✔ ANSWER

Observation: A **white precipitate** (insoluble in water) is formed.

Conclusion: The precipitate is barium sulphate; ions are exchanged (**double displacement / precipitation reaction**).

$Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$

(i) **Recall Activity 1.2 (lead(II) nitrate + potassium iodide). What was the colour of the precipitate formed? Can you name the compound precipitated?**

✔ ANSWER

The precipitate is **yellow**. It is **lead(II) iodide, PbI_2.**

(ii) **Write the balanced chemical equation for this reaction.**

✔ ANSWER

$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$

(iii) **Is this also a double displacement reaction?**

✔ ANSWER

Yes. Lead and potassium exchange their ions (I^- and NO_3^-). It is also a precipitation reaction.

⭐ 1.2.5 Oxidation and Reduction

Activity 1.11 **Heat a china dish containing about 1 g copper powder.**

What do you observe?

✔ ANSWER

Observation: The brown surface of copper powder becomes coated with a **black** substance.

Conclusion: Copper gains oxygen and forms black **copper(II) oxide**: copper is **oxidised**.

$2Cu + O_2 \rightarrow (Heat)\ 2CuO$

? **Recall Activity 1.1, where a magnesium ribbon burns in air (oxygen) and changes into magnesium oxide. Is magnesium being oxidised or reduced in this reaction?**

✔ ANSWER

Magnesium **gains oxygen** to form MgO, so magnesium is being **oxidised**.

$2Mg + O_2 \rightarrow 2MgO$

? Questions · 1.2.3–1.3 Displacement, Redox, Corrosion and Rancidity

Q 1 **Why does the colour of copper sulphate solution change when an iron nail is dipped in it?**

✔ ANSWER

Iron is more reactive than copper, so it displaces copper from copper sulphate solution.

Blue copper sulphate changes into **pale green iron sulphate**, so the blue colour fades.

$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$

Q 2 **Give an example of a double displacement reaction other than the one given in Activity 1.10.**

✔ ANSWER

Lead nitrate and potassium iodide:

$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$
(Another: $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$)

Q 3 (i) **Identify the substances oxidised and reduced: $4Na(s) + O_2(g) \rightarrow 2Na_2O(s)$**

✔ ANSWER

Oxidised: sodium (Na) – it gains oxygen.

Reduced: oxygen (O_2) – it combines with sodium (it is the oxidising agent).

Q 3 (ii) **$CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(l)$**

✔ ANSWER

Oxidised: hydrogen (H_2) – it gains oxygen to form H_2O.

Reduced: copper(II) oxide (CuO) – it loses oxygen to form Cu.

📝 Exercises

1 **Which of the statements about the reaction below are incorrect?**

$2PbO(s) + C(s) \rightarrow 2Pb(s) + CO_2(g)$

(a) Lead is getting reduced. (b) Carbon dioxide is getting oxidised. (c) Carbon is getting oxidised. (d) Lead oxide is getting reduced.

(i) (a) and (b) (ii) (a) and (c) (iii) (a), (b) and (c) (iv) all

✔ ANSWER

(i) (a) and (b)

Lead oxide (not lead) is reduced, and carbon dioxide is a product, not something being oxidised. (c) and (d) are correct.

2 $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$

The above reaction is an example of a (a) combination reaction (b) double displacement reaction (c) decomposition reaction (d) displacement reaction.

✔ ANSWER

(d) displacement reaction

Aluminium (more reactive) displaces iron from iron oxide.

3 **What happens when dilute hydrochloric acid is added to iron fillings? Tick the correct answer.**

(a) Hydrogen gas and iron chloride are produced. (b) Chlorine gas and iron hydroxide are produced. (c) No reaction takes place. (d) Iron salt and water are produced.

✔ ANSWER

(a) Hydrogen gas and iron chloride are produced.

$Fe(s) + 2HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)$

4 **What is a balanced chemical equation? Why should chemical equations be balanced?**

✔ ANSWER

A **balanced chemical equation** has the same number of atoms of each element on the reactant side (LHS) and the product side (RHS).

Example: $2H_2 + O_2 \rightarrow 2H_2O$

Equations must be balanced because of the **law of conservation of mass**: mass can neither be created nor destroyed in a chemical reaction, so the total mass (number of atoms of each element) must be the same before and after the reaction.

5 (a) **Translate into a chemical equation and balance: Hydrogen gas combines with nitrogen to form ammonia.**

✔ ANSWER

$3H_2(g) + N_2(g) \rightarrow 2NH_3(g)$

5 (b) **Hydrogen sulphide gas burns in air to give water and sulphur dioxide.**

✔ ANSWER

$2H_2S(g) + 3O_2(g) \rightarrow 2H_2O(l) + 2SO_2(g)$

5 (c) Barium chloride reacts with aluminium sulphate to give aluminium chloride and a precipitate of barium sulphate.

✔ ANSWER

$3BaCl_2(aq) + Al_2(SO_4)_3(aq) \rightarrow 2AlCl_3(aq) + 3BaSO_4(s)$

5 (d) Potassium metal reacts with water to give potassium hydroxide and hydrogen gas.

✔ ANSWER

$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$

6 (a) Balance: $HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + H_2O$

✔ ANSWER

$2HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + 2H_2O$

6 (b) $NaOH + H_2SO_4 \rightarrow Na_2SO_4 + H_2O$

✔ ANSWER

$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

6 (c) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

✔ ANSWER

$NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
(Already balanced: each atom is 1 on both sides.)

6 (d) $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + HCl$

✔ ANSWER
$BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$

7 (a) Write the balanced equation: Calcium hydroxide + Carbon dioxide → Calcium carbonate + Water

✔ ANSWER
$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

7 (b) Zinc + Silver nitrate → Zinc nitrate + Silver

✔ ANSWER
$Zn + 2AgNO_3 \rightarrow Zn(NO_3)_2 + 2Ag$

7 (c) Aluminium + Copper chloride → Aluminium chloride + Copper

✔ ANSWER
$2Al + 3CuCl_2 \rightarrow 2AlCl_3 + 3Cu$

7 (d) Barium chloride + Potassium sulphate → Barium sulphate + Potassium chloride

✔ ANSWER

$BaCl_2 + K_2SO_4 \rightarrow BaSO_4 + 2KCl$

8 (a) Write the balanced equation and identify the type: Potassium bromide(aq) + Barium iodide(aq) → Potassium iodide(aq) + Barium bromide(s)

✔ ANSWER

$2KBr(aq) + BaI_2(aq) \rightarrow 2KI(aq) + BaBr_2(s)$

Type: **double displacement** (precipitation) reaction.

8 (b) Zinc carbonate(s) → Zinc oxide(s) + Carbon dioxide(g)

✔ ANSWER

$ZnCO_3(s) \rightarrow^{(Heat)} ZnO(s) + CO_2(g)$

Type: **decomposition** (thermal decomposition) reaction.

8 (c) Hydrogen(g) + Chlorine(g) → Hydrogen chloride(g)

✔ ANSWER

$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

Type: **combination** reaction.

8 (d) **Magnesium(s) + Hydrochloric acid(aq) → Magnesium chloride(aq) + Hydrogen(g)**

✔ ANSWER

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

Type: **displacement** reaction.

9 **What does one mean by exothermic and endothermic reactions? Give examples.**

✔ ANSWER

Exothermic reaction: heat is released along with the formation of products.

- $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$
- $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$ (burning of natural gas)

Endothermic reaction: energy (heat, light or electricity) is absorbed.

- $CaCO_3(s) \rightarrow (Heat)\ CaO(s) + CO_2(g)$
- $2AgCl(s) \rightarrow (Sunlight)\ 2Ag(s) + Cl_2(g)$

10 **Why is respiration considered an exothermic reaction? Explain.**

✔ ANSWER

During digestion, carbohydrates in food are broken down into glucose.

In our cells, glucose combines with oxygen and gives carbon dioxide, water and **energy**.

$C_6H_{12}O_6(aq) + 6O_2(aq) \rightarrow 6CO_2(aq) + 6H_2O(l) +$ energy

Since energy is released, respiration is an **exothermic** reaction.

11 **Why are decomposition reactions called the opposite of combination reactions? Write equations for these reactions.**

✔ ANSWER

In a **combination** reaction, two or more substances combine to form a **single product**.

In a **decomposition** reaction, a **single substance** breaks down into two or more simpler substances. So one is the reverse of the other.

- Combination: $CaO(s) + CO_2(g) \rightarrow CaCO_3(s)$; $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$
- Decomposition: $CaCO_3(s) \rightarrow$(Heat) $CaO(s) + CO_2(g)$; $2H_2O(l) \rightarrow$(Electricity) $2H_2(g) + O_2(g)$

Also, combination reactions usually release energy, while decomposition reactions need energy.

12 **Write one equation each for decomposition reactions where energy is supplied in the form of heat, light or electricity.**

✔ ANSWER

Heat: $CaCO_3(s) \rightarrow (Heat)\ CaO(s) + CO_2(g)$

Light: $2AgCl(s) \rightarrow (Sunlight)\ 2Ag(s) + Cl_2(g)$

Electricity: $2H_2O(l) \rightarrow (Electricity)\ 2H_2(g) + O_2(g)$

13 **What is the difference between displacement and double displacement reactions? Write equations for these reactions.**

✔ ANSWER

Displacement	Double displacement
A more reactive element displaces a less reactive element from its compound	Two compounds exchange their ions
One element and one compound react	Two compounds react
$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$	$Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$

14 **In the refining of silver, the recovery of silver from silver nitrate solution involved displacement by copper metal. Write down the reaction involved.**

✔ ANSWER

Copper is more reactive than silver, so it displaces silver:

$Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$

15 **What do you mean by a precipitation reaction? Explain by giving examples.**

✔ ANSWER

A reaction in which an **insoluble solid (precipitate)** is formed is called a precipitation reaction.

- $Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$ – white precipitate of barium sulphate.
- $Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$ – yellow precipitate of lead iodide.

These are also double displacement reactions.

16 (a) **Explain oxidation in terms of gain or loss of oxygen with two examples.**

✔ ANSWER

Oxidation is the **gain of oxygen** by a substance.

- $2Cu + O_2 \rightarrow$(Heat) $2CuO$ (copper is oxidised)
- $C + O_2 \rightarrow CO_2$ (carbon is oxidised)

16 (b) **Explain reduction in terms of gain or loss of oxygen with two examples.**

✔ ANSWER

Reduction is the **loss of oxygen** by a substance.

- $CuO + H_2 \rightarrow$(Heat) $Cu + H_2O$ (copper oxide is reduced)
- $ZnO + C \rightarrow Zn + CO$ (zinc oxide is reduced)

17 **A shiny brown coloured element 'X' on heating in air becomes black in colour. Name the element 'X' and the black coloured compound formed.**

✔ ANSWER

'X' is **copper (Cu)**. The black compound is **copper(II) oxide (CuO)**.

$2Cu + O_2 \rightarrow$(Heat) $2CuO$

18 **Why do we apply paint on iron articles?**

✔ ANSWER

Paint forms a layer that keeps **air (oxygen) and moisture** away from the iron surface.

This prevents **rusting (corrosion)** of iron articles.

19 **Oil and fat containing food items are flushed with nitrogen. Why?**

✔ ANSWER

Nitrogen is an unreactive gas. It removes oxygen (air) from the packet.

This prevents **oxidation** of the oils and fats, so the food does not become **rancid** (bad in smell and taste).

20 (a) **Explain with one example: Corrosion**

✔ ANSWER

When a metal is attacked by substances around it, such as moisture, air and acids, it slowly gets damaged. This is **corrosion**.

Example: **rusting of iron** – iron gets coated with a reddish-brown powder (rust). (Black coating on silver and green coating on copper are also corrosion.)

20 (b) **Explain with one example: Rancidity**

✔ ANSWER

When fats and oils in food are **oxidised**, their smell and taste change. This is **rancidity**.

Example: chips or fried snacks left open for a long time taste and smell bad.

Prevented by adding antioxidants, air-tight containers, or flushing packets with nitrogen.

🧪 Group Activity

? **Add potassium sulphate, ammonium nitrate, anhydrous copper sulphate to water in beakers A, B, C and fine iron fillings to copper**

sulphate solution in beaker D. Record temperatures before and after.

Find out which reactions are exothermic and which ones are endothermic in nature.

💡 **SAMPLE ANSWER**

Beaker	Change in temperature	Nature
A – potassium sulphate in water	falls slightly	endothermic
B – ammonium nitrate in water	falls clearly (feels cold)	endothermic
C – anhydrous copper sulphate in water	rises	exothermic
D – iron fillings in copper sulphate solution	rises	exothermic

A rise in temperature means heat is given out (exothermic); a fall means heat is absorbed (endothermic). (Record your own readings.)

🎉 **Chapter 1 done! Revise the answers once more tomorrow.**